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LEXFLIP: A Dissociation Diagnostic for Legal Meaning Preservation Metrics

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Abstract: Does a simplified legal clause still say what the original said? The checks in current use cannot establish that it does: requiring an identical pair to score highest and an unrelated pair lowest moves lexical overlap and legal force together, so any monotone function of token overlap satisfies both. Our remedy is a *dissociation*, an item holding surface form fixed while legal force moves. We release LEXFLIP, 373 minimal perturbations of Quebec statutory French that reverse legal force while preserving 0.93 of the tokens, with a harness scoring metrics, regressors and prompted judges alike. The seven embedding and BERTScore metrics we test spend only 0.022–0.039 of their identical-to-unrelated range on such an edit, against 0.670 for bidirectional NLI, the one family the identical-pair check would disqualify. On FRJUDGE, against a measured human ceiling of $r = 0.597$, a bare length feature outscores every semantic metric and has the lowest margin we measure.

Keywords: legal text simplification, meaning preservation, evaluation metrics, construct validity, natural language inference, large language model judges, French legal language

1. Introduction

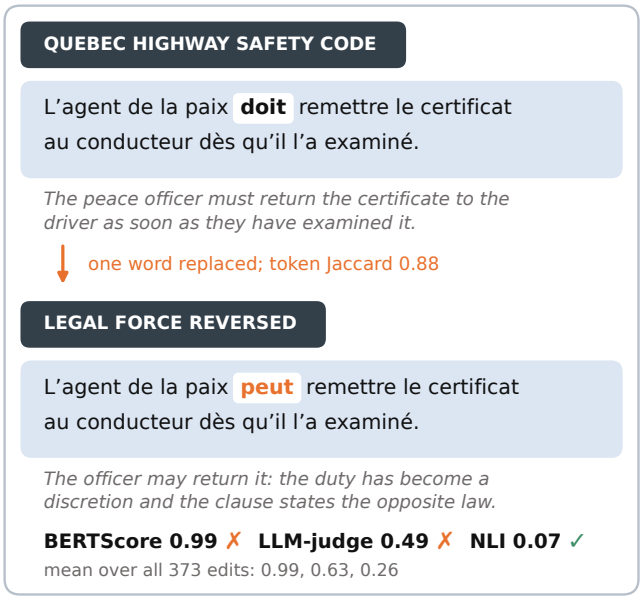
Rewriting a legal text for a lay reader has an unusual failure mode. When a news summary drops a qualifier the reader is mildly misinformed; when an insurance clause drops a qualifier the reader may be uninsured. *Moffatt v. Air Canada* (2024) held an airline liable for its chatbot's misstatement of its own policy, and legal research tools sold as hallucination-free still err on 17 to 33% of queries [1].

If simplified legal text is produced automatically, an automatic metric is what checks it. Two checks are in current use [2, 3]: a sentence paired with itself must receive the maximum score, and a sentence paired with an unrelated one the minimum. Both are necessary and both are easy to state. But moving from an identical pair to an unrelated pair changes lexical overlap and legal meaning *together*, in the same direction, at the same time, so any monotone function of token overlap satisfies both by construction. The checks test calibration at two endpoints; they do not test what the metric is reading.

The remedy is a *dissociation*: an item on which the two confounded variables are pulled apart, so that legal force changes while surface form does not. Replace *doit* (must) with *peut* (may) and ninety-odd percent of the tokens are untouched, the sentence stays fluent and statutory in register, and an obligation has become a permission (Figure 1). A metric that measures legal meaning must drop sharply; a metric that measures overlap cannot. We release LEXFLIP, 373 such perturbations of Quebec statutory French, with a harness taking an arbitrary scoring function so that untrained metrics, regressors and prompted judges take the same test,[1] and a four-part validation protocol around it.

This article makes four contributions. First, it states a four-part validation protocol for any scorer of legal meaning preservation, phrased over an arbitrary scoring function rather than over an architecture. Second, it releases LEXFLIP, a diagnostic set of 373 minimally edited Quebec statutory sentences whose legal force is reversed and whose surface form is almost unchanged. Third, it administers the protocol to thirteen scorers, including untrained similarity metrics, fine-tuned regressors and a prompted large language model judge, and shows that correlation with human ratings and sensitivity to a legally decisive edit come apart. Fourth, it reports what happens when a metric is trained against the protocol, and finds

Figure 1. One LEXFLIP item. Replacing *doit* with *peut* turns a duty into a discretion and leaves the sentence otherwise untouched, so a metric reading legal meaning must drop and one reading overlap cannot. The judge's ten-point rating is rescaled to $[0, 1]$, as in Table 2. Over the whole set the edit costs BERTScore 0.022 of its own identical-to-unrelated range and the judge 0.370, against 0.670 for bidirectional NLI.



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[1] https://github.com/Nyvora-Vision-Labs/LEXFLIP

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that the constraint is satisfied in distribution and not at all across a change of source document.

1.1. Related Work

BLEU does not correlate with meaning preservation once sentence splitting is involved [4], and BERTScore [5] and other embedding metrics score high across *all* alterations. FRJUDGE [3] is to our knowledge the only annotated resource for legal meaning preservation in any language, and trained metrics fitted to such corpora [2] are our point of departure. Closest in method, Chen and Eger [6] show similarity metrics are not robust to meaning-changing edits while metrics based on natural language inference repair much of the gap, and Mujahid et al. [7] run the complementary experiment on factual consistency in summarization. No prior challenge set we know of targets legal force, which is the property a simplification of a statute or an insurance clause has to preserve and the one a lay reader cannot check. We follow Bean et al. [8] and Pacchiardi et al. [9] on construct validity, Deutsch et al. [10] and Graham and Baldwin [11] on statistics, and Xu and Jurgens [12] on rating distributions.

2. What Would Validate a Legal Meaning Metric?

The four requirements below are stated for an arbitrary scorer $m(o,s)$ over an original clause o and a candidate simplification s; none presupposes an architecture.

R1: a measured ceiling. A correlation with human judgment means nothing until one knows what correlation a human achieves. Where a corpus carries $k \geq 3$ ratings per item this is free: hold out one rater, correlate their ratings against the mean of the other $k-1$, average over the held-out position. That answers *is the metric as good as an expert?*; the Spearman–Brown reliability of the k-rater mean answers *is it as good as the aggregate label allows?* A metric can sit above the first and below the second without being superhuman, since predicting an average is easier than being a rater.

R2: identical supervision. Comparisons here place an unsupervised cosine similarity and a fine-tuned regressor in one table, decimal-scale the former, and report RMSE for both. Decimal scaling fixes the range but neither the location nor the shape of the distribution, so a similarity occupying $[0.7, 1.0]$ records a large RMSE however well it ranks, and the column measures supervision rather than judgment. We instead fit a monotone map from raw score to human scale on the training split alone; isotonic regression cannot improve a ranking, and rank statistics move by at most 0.014.

R3: a trivial-feature control. Simplifications are usually shorter and the dominant annotated error category is omission, so a rubric deducting score once per identified error yields a label partly predictable from how many words disappeared. We require a supervised baseline over surface features alone, fitted under R2.

R4: dissociation. On naturally occurring pairs surface form and legal force vary together, the confound the endpoint checks inherit. Let s' be a perturbation of s with token Jaccard $J(s,s') \approx 1$ and legal force $L(s') \neq L(s)$, and u an unrelated sentence of the same register. We report the *margin fraction* $(\overline{m(s,s)} - \overline{m(s,s')})/(\overline{m(s,s)} - \overline{m(s,u)})$, the share of the metric's own identical-to-unrelated range spent on a legally decisive edit, which charges it against the range it actually has rather than a nominal $[0,1]$. A metric treating the flipped sentence as it treats an unrelated one scores 1; a pure overlap function scores near 0. This is not the rate at which (s,s) outranks (s,s'), the weaker quantity: a metric can rank every item correctly while moving a thousandth of its range.

Table 1. LEXFLIP samples, one per perturbation rule, with the edited span in bold. Each remains fluent and statutory in register, shares almost all of its tokens with the original, and states a different rule of law.

peut → *doit*	*tier 1*
o: La Société peut refuser une indemnité, en réduire le montant, en suspendre ou en cesser le paiement dans les cas suivants: 1.	
s′: La Société **doit** refuser une indemnité, en réduire le montant, en suspendre ou en cesser le paiement dans les cas suivants: 1.	
doit → *peut*	*tier 1*
o: Le candidat doit toutefois réussir les examens de compétence visés à l'article 67 pour obtenir un permis de conduire pour motocyclette.	
s′: Le candidat **peut** toutefois réussir les examens de compétence visés à l'article 67 pour obtenir un permis de conduire pour motocyclette.	
drop *ne . . . pas*	*tier 1*
o: Les feux visés au paragraphe 2° ne sont pas requis pour tout véhicule dont la largeur excède 2,03 mètres.	
s′: Les feux visés au paragraphe 2° sont requis pour tout véhicule dont la largeur excède 2,03 mètres.	
avant ↔ *après*	*tier 1*
o: Elle n'est pas due avant le septième jour qui suit celui de l'accident, sauf dans le cas prévu au troisième alinéa de l'article 57.	
s′: Elle n'est pas due **après** le septième jour qui suit celui de l'accident, sauf dans le cas prévu au troisième alinéa de l'article 57.	
tous les → *certains*	*tier 1*
o: Un agent de la paix peut enlever ou faire enlever aux frais du propriétaire toute chose utilisée en contravention au présent article.	
s′: Un agent de la paix peut enlever ou faire enlever aux frais du propriétaire **certaine** chose utilisée en contravention au présent article.	

3. Materials and Methods

3.1. The LEXFLIP Diagnostic

Source sentences come from two Quebec statutes held out from every corpus used here, the Automobile Insurance Act and the Highway Safety Code, and each takes one minimal edit that changes legal force while leaving the sentence intact. The inventory is organised by legal effect, not by linguistic category. Six families make up *tier 1*, where the edit unambiguously changes who is bound, what is covered or how much: modality (*doit* ↔ *peut*), party (*l'assuré* ↔ *l'assureur*), polarity (dropping *ne . . . pas*), quantum (×10), scope (*tous les* → *certains*) and temporal (*avant* ↔ *après*); *tier 2* connective and class shifts are reported separately. The diagnostic would fail if a metric could find the edited item by noticing that it reads badly, so perturbations introducing a grammatical artefact are filtered by a conservative French well-formedness check that flags none of the 400 unmodified sources. The result is 373 pairs (310 tier 1) with mean token Jaccard 0.93, none below 0.75. Table 1 gives one sample per rule.

For each s with perturbation s' we score (s,s), (s,s') and (s,u) for an unrelated u from the same statutes, min–max normalise per metric so the two probes anchor the range, and take the margin from the three means.

3.2. Scorers Evaluated

Only French-capable metrics are administered, since scoring English-only ones on French confounds coverage with quality. The

set covers three families, named in Table 2: BERTScore [5] over CamemBERTv2 and FlauBERT, in F1 and recall-only variants since recall targets omission; four sentence-embedding cosines (LaBSE, multilingual E5-base, paraphrase-multilingual-mpnet and Sentence-CamemBERT); and bidirectional natural language inference with mDeBERTa-XNLI [13], where failure of $o \Rightarrow s$ signals hallucination and of $s \Rightarrow o$ omission, the two dominant categories of the legal error taxonomy. We add token Jaccard and the surface features of R3.

3.3. The FRJUDGE Corpus and Its Reconstruction

The diagnostic answers R4. We run the other three on FRJUDGE [3], 297 insurance clauses simplified by `gpt-4-turbo-2024-04-09` and rated by five law students on legal meaning (1–10), then ask whether the two kinds of evidence agree. The public release drops annotator identity, making R1 impossible downstream, so we rebuild the corpus from the raw Prodigy export, recovering the five annotators (pseudonymised A–E) and reproducing the published pairwise agreement of 25.96%. The same export carries the source document of each clause, which the public release also drops and which Section 4.7 uses to build a document-grouped split.

Splits are 178/30/89 pairs for training, development and test, drawn under ten seeds for the unsupervised comparison and five for the trained models. Isotonic calibration under R2 is fitted on the training split alone and never on the split a number is reported from.

3.4. A Prompted Judge

A prompted model given the annotators' rubric [15] is the natural competitor to a fine-tuned 112M-parameter encoder. We score each pair with `deepseek-chat` under the FRJUDGE scale, a ten-point judgment deducting one point per identified legal error, in its own request so the judge never sees the probe condition, $K = 5$ times, and average the samples. Three prompt wordings are compared: the FRJUDGE rubric in French, the same rubric in English over the same French text, and a bare instruction with the four-type error taxonomy removed. A reasoning model (`deepseek-reasoner`) is run under the identical rubric and items as a capability control. The full protocol is 7,080 calls costing $0.99.

3.5. Trained Metric Variants

The protocol so far audits scorers that already exist. It also has implications for how one would fit a new one, and we test two of them. The first follows from R1: the gold label is the mean of five ratings whose median spread is 7 points, so fitting a scalar regressor to that mean asks the model to predict a quantity none of the annotators asserted, and rewards predicting the centre. The second follows from R4: if a diagnostic identifies a deficiency, the obvious response is to train against it.

The seven configurations share an encoder (CamemBERTv2-base), splits, schedule and seeds, and differ only in the target and the loss.

- **Scalar** (the conventional objective): mean squared error against the mean rating.
- **Distributional**: a ten-way head trained with KL divergence against the empirical rating histogram, following Uma et al. [16] and Peterson et al. [17]. The reported score is the distribution's expectation, so it is directly comparable, but the model additionally exposes its spread.
- **Annotator-aware**: five regression heads, one per annotator, averaged at inference. This lets the model represent the lenient and strict populations of Section 4.3 separately instead of averaging them away.
- **Quantile**: a pinball loss [18] at $\tau = 0.25$. The asymmetry is deliberate. A metric that *overshoots* the human rating is permissive of bad simplifications, while one that undershoots is merely strict, and in a legal setting those errors are not symmetric; a quantile objective encodes that preference in the loss rather than hoping it emerges.
- **Multi-task**: the scalar head plus an auxiliary characterization head, trained against the majority vote of the five annotators, which spans 17 of the rubric's 18 classes (the eighteenth never carries a majority). The characterization labels are collected in FRJUDGE and then discarded; since the published rubric has annotators characterize a clause *before* scoring it, the label plausibly carries information about the score.
- **DA**: the scalar objective with the identical/unrelated augmentation in current use added to the training mixture, unchanged.
- **DA+LexFlip**: **DA** plus minimally-edited negatives generated from FRJUDGE originals by the rules of Section 3.1, disjoint from the diagnostic's own sentences.

Correlation and RMSE are computed *only* on the human-annotated test split, never on a split containing augmentation. This matters: adding identical pairs (label 10) and unrelated pairs (label 1) to the evaluation set inflates label variance at the extremes, which raises Pearson r for every metric without any improvement in judgment. A table that does not separate the two shows the signature clearly, in that a metric's r and its RMSE improve and worsen together rather than moving as one. Augmentation here affects training only, and the three diagnostic regimes are reported separately as pass rates.

3.6. Statistical Procedures

The test split holds 89 items, at which size the smallest correlation reliably distinguishable from $r = 0.46$ at 80% power is 0.73. The unsupervised metrics and the length feature correlate with the *same* human label on the *same* 297 items, and with each other, so a two-sample test on Fisher-transformed correlations would be anticonservative; we use the Williams test [11], which corrects for that dependence. Confidence intervals on correlations are bootstrap intervals over items. Trained models are reported as means and standard deviations over five seeds and unsupervised comparisons over ten, and we treat any difference smaller than that spread as unresolved.

Every number we report is produced by a single pipeline that runs end to end on a laptop; the largest model involved is CamemBERTv2-base (112M parameters) and no GPU cluster is required. Training runs use Apple MPS kernels, which are not bitwise deterministic, so a fixed seed reproduces a run only approximately: repeat runs of the scalar configuration at seed 42 differ in test-set r by several hundredths, comparable to the spread across seeds. Anyone reproducing this on CUDA should expect tighter agreement and correspondingly narrower error bars.

3.7. Use of Generative AI

Generative AI enters this work in two distinct roles, and we separate them because only one is a result. As an

Table 2. LexFlip, min–max normalised per metric. *discrim.* is the share of items ranked correctly (identical > edited), *margin* the fraction of the identical-to-unrelated range spent on the legal edit.

Metric	identical	LexFlip	unrelated	discrim.%	margin
NLI-bidirectional (min)	0.784	0.265	0.010	93.3	0.670
NLI-bwd (no omission)	0.784	0.432	0.042	77.2	0.473
NLI-fwd (no hallucination)	0.784	0.469	0.043	77.7	0.424
LLM-judge (deepseek-chat, k=5)	1.000	0.632	0.006	86.3	0.370
surface: token Jaccard	1.000	0.928	0.089	99.2	0.079
LaBSE	1.000	0.972	0.292	100.0	0.039
BERTScore-FlauBERT	1.000	0.979	0.393	100.0	0.035
mE5-base	1.000	0.975	0.266	100.0	0.034
Para-mUSE-mpnet	1.000	0.981	0.312	100.0	0.028
Sentence-CamemBERT	1.000	0.983	0.316	100.0	0.024
BERTScore-CamemBERTv2-Recall	1.000	0.983	0.240	100.0	0.023
BERTScore-CamemBERTv2	1.000	0.982	0.193	100.0	0.022
surface: $-\lvert\lvert s\rvert - \lvert o\rvert\rvert$	1.000	0.997	0.718	11.3	0.011

object of study, `deepseek-chat` and `deepseek-reasoner` are scored as prompted judges under the protocol of Section 3.4, and `gpt-4-turbo-2024-04-09` produced the simplifications in FrJudge before this work began. Separately, a large language model assistant was used to rephrase passages of the manuscript for clarity. No AI system is an author. The author designed the study, verified every reported number against the released result files, and takes full responsibility for the content of this article, including any part produced with the help of such a tool.

4. Results

4.1. Similarity Metrics Fail the Dissociation

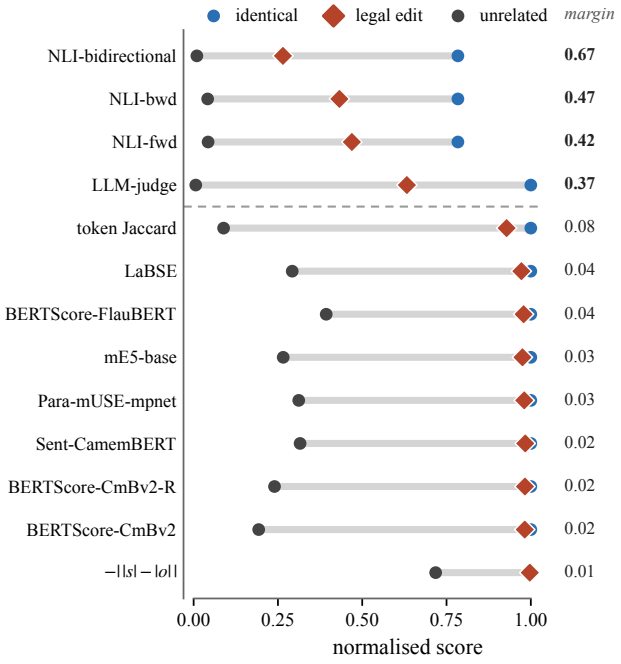
Every *similarity*-based metric fails, and fails almost completely (Table 2). The seven embedding and BERTScore variants spend between 0.022 and 0.039 of their working range on an edit that reverses the law: BERTScore over CamemBERTv2 places a sentence saying the opposite of the original at 0.982, where the original sits at 1.000 and an unrelated statute at 0.193. Token Jaccard, at 0.079, is in the same regime, which is the point: these are overlap functions, exactly as Section 2 says they must be. Their *discrimination* rates are uninformative for the same reason, the seven ranking the identical pair first 100% of the time by a margin that carries no decision. Bidirectional NLI does not fail: it spends 0.670 of its range on the edit, ranks correctly on 93.3% of items, and places a flipped sentence nearer the unrelated pair than the identical one. Figure 2 shows the same result as a picture: for every similarity metric the flipped sentence sits hard against the identical pair, at the wrong end of the metric's own range.

The existing checks would have excluded it. Note where NLI's identical score sits: 0.784, and not as a normalisation artefact. In raw terms mDeBERTa assigns a mean entailment of 0.780 to a sentence paired with *itself*, clearing the 99% bar the identical-pair convention requires on only 1.3% of items, against 100% for every BERTScore and embedding metric. The metric that best tracks legal meaning is the one convention would *disqualify*. Saturation on identical input is easy to build, every overlap function having it by construction: report it, but do not filter on it.

4.2. Which Edits Are Detected

No perturbation rule is reliably detected (Table 3). Averaged over the 13 scorers the best rule, dropping *ne . . . pas*, reaches 0.333, and the party swap between insurer and insured reaches 0.035. Two details behind that party-swap figure are worth stating. The two

Figure 2. LexFlip. For each metric the grey bar spans its score on an unrelated pair (dark circle) to its score on an identical pair (blue circle): its full working range. The red diamond is where it places a sentence whose legal force has been reversed by a single word. A metric that measured legal meaning would put the diamond at the left end. Every similarity-based metric puts it at the right, hard against the identical pair. Only the four scorers above the dashed rule, the three NLI variants and the prompted judge, move it appreciably. Metric names are abbreviated; Table 2 gives them in full, with the margin column repeated at the right here.



directions of the swap draw mean margins of 0.046 and −0.005 on 11 and 3 items respectively, which is why we pool them rather than report either cell on its own: over all 14 party-swap items the seven similarity metrics spend 0.020 of their range, with a per-metric spread of 0.006–0.036, against 0.029 for those same seven metrics across all rules. The contrast is milder on this restricted basis than the all-metric figures suggest, because the similarity family has almost no range to spend on any rule.

Table 3. LEXFLIP margin by perturbation rule, averaged over the metrics in Table 2. *worst* is the lowest margin any single metric assigns that rule. A margin of 1 would mean the metric treats a legally flipped sentence exactly as it treats an unrelated one.

Perturbation	n	mean margin	worst
drop *ne . . . pas*	21	0.333	0.032
avant ↔ *après*	9	0.272	0.006
amount × 10	21	0.256	0.025
doit → *peut*	82	0.241	0.015
sauf si → *si*	10	0.225	0.014
peut → *doit*	111	0.167	0.018
tous les → *certains*	42	0.153	0.017
et → *ou*	27	0.069	0.007
ou → *et*	26	0.069	0.004
assureur → *assuré*	11	0.046	0.007
automobile ↔ *véhicule*	10	0.027	-0.025
assuré → *assureur*	3	-0.005	-0.086
All	373	**0.155**	–

Table 4. Krippendorff's α on FRJUDGE. Legal meaning is an ordinal ten-point scale, for which the nominal coefficient is not the appropriate measure. *prev.* is the value reported by Beauchemin et al. [3].

Dimension	Krippendorff's α		Reported
	nominal	appropriate	prev.
Simplicity level	0.183	0.183	0.18
Characterization	0.552	0.552	0.55
Legal meaning	0.104	**0.325**	0.10

We also tested whether the residual ordering across rules is explained by the size of the edit in characters, and found no relationship ($r = 0.25$ over the 12 rules, 95% CI $[-0.66, 0.73]$). With 12 rules of unequal frequency the ordering is not itself well determined, so we report it descriptively; the robust observation is the ceiling on detectability, not the ranking.

4.3. Disagreement and the Human Ceiling

Krippendorff's α for legal meaning is 0.104 under the *nominal* coefficient, the one conventionally reported here, and 0.325 under the *ordinal* one appropriate to an ordered scale [14]; Table 4 gives all three annotation dimensions, and our nominal figures reproduce the published ones. The ordinal coefficient is computed with the same library and the same reliability matrix as the nominal one, so the difference between 0.104 and 0.325 is attributable to the distance function alone.

What remains after that correction is structural: the annotators separate into two stable populations, B, C and D averaging 7.61 against 4.34 for A and E, and 63% of pairs span at least 6 points (Figure 3). Averaging leaves a target with standard deviation 2.40, the condition under which a regressor predicting the middle scores well. Correlating each annotator against the mean of the other four gives a ceiling of 0.597 (RMSE 3.27), with the held-out annotators ranging from 0.524 to 0.680 (Table 5); exceeding that ceiling is not superhuman, the bound for predicting that mean being its own reliability, 0.811 by Spearman–Brown, so a metric can approach $\sqrt{0.811} = 0.90$.

Table 5. Leave-one-annotator-out ceiling on FRJUDGE ($n = 297$). A single expert predicts the consensus of the other four at $r = 0.597$, and the best-agreeing pair of annotators reaches $r = 0.606$.

Held-out	r (95% CI)	ρ	RMSE
A	0.524 [0.44, 0.60]	0.551	3.93
B	0.569 [0.48, 0.65]	0.586	3.05
C	0.680 [0.60, 0.75]	0.725	2.75
D	0.601 [0.52, 0.67]	0.557	3.08
E	0.612 [0.55, 0.67]	0.679	3.56
Mean (ceiling)	**0.597**	0.620	3.27
Best pair (C-D)	0.606	–	–
Mean pairwise	0.462	–	–

4.4. Identical Supervision, and a Trivial Control

Calibration collapses most of the RMSE gap uncalibrated comparisons record (Table 6): the metrics were not as far from human judgment as such tables suggest, merely on a different scale. And a ridge over word counts and token overlap reaches $r = 0.62$, at the human ceiling and above every semantic metric we test, the best reaching $r = 0.482$. Only two scorers beat it, and neither is semantic: the prompted judge below, and the bare length difference at $r = 0.641$. Under the Williams test every one of the ten unsupervised neural metrics is significantly *worse* than that length feature (Table 7: t between 2.61 and 6.33, $p \leq 0.010$), so the ordering is not an artefact of sampling. This says less about the corpus than about what the *annotation* rewards, score being reduced once per identified error and the most frequent error type, omission, correlating strongly with becoming shorter.

4.5. The Prompted Judge

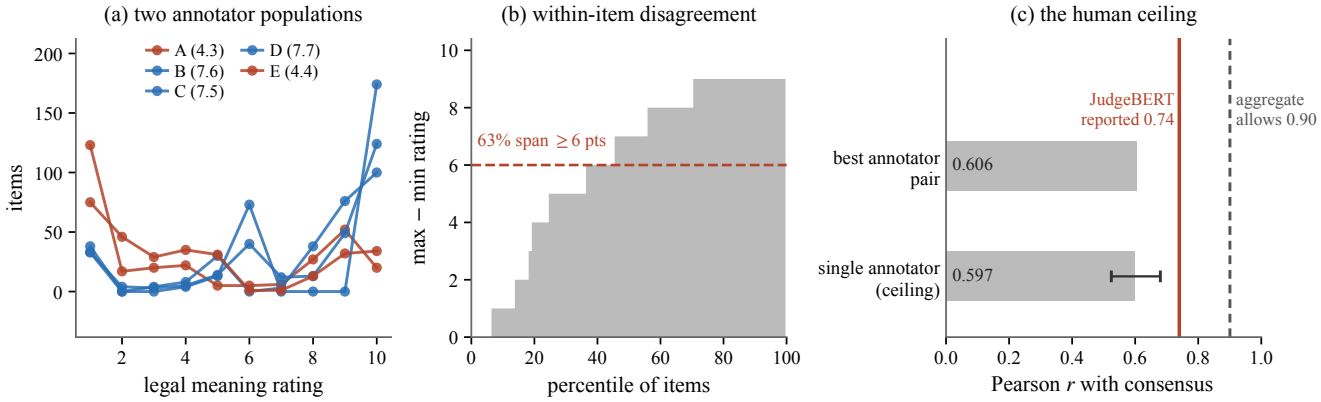
Under R2 the prompted judge reaches $r = 0.755_{\pm 0.054}$, above the length feature and our ceiling (a redraw gave 0.726). On LEXFLIP it scores the identical pair at 10.00 and the unrelated pair at 1.06, flawless on the two checks jointly satisfiable by overlap, then awards a clause whose legal force has been reversed 6.69 out of ten, a margin of 0.370.

The rubric, not the model, limits this. With the model fixed the margin moves from 0.421 under the rubric-faithful prompt to 0.687 under a bare instruction stripping the four-type taxonomy, against 0.002 for a redraw, and the correlation moves the other way, from 0.687 to 0.758. Score is reduced *once per identified error*, so a clause reversal is one incoherence and the penalty is capped near the top of the scale. Substituting a reasoning model under the identical rubric and items does not help: `deepseek-reasoner` gives $r = 0.580$ against 0.619 and margin 0.320 against 0.425 at matched sampling, so the shortfall is not a matter of the model thinking harder.

One behaviour survives every configuration: *peut* for *doit* scores 4.72 while the reverse scores 8.02, though both change the law by the same construction. The gap ranges from 3.24 to 4.03 across the 5 configurations we ran. The judge therefore tracks legal *severity* rather than legal *equivalence*, penalising a clause that becomes more permissive and forgiving one that becomes more demanding.

The judge as a sixth annotator. Because the judge emits on the annotators' scale it can be placed inside the panel rather than only beside the metrics. Its mean rating is 9.11, against a human panel

Figure 3. Disagreement in FRJUDGE. (a) The five annotators form two populations: B, C, D (blue) rate legal preservation about three points higher than A and E (red) across the whole corpus. (b) Sorted within-item range; the median item spans 7 points of a ten-point scale. (c) A single annotator predicts the consensus of the other four at $r = 0.597$ (bar; whiskers give the range over the five held-out annotators), and the best-agreeing pair reaches $r = 0.606$.



spanning 4.34 to 7.61, so it is more lenient than any human and far less dispersed. Yet its mean correlation with the five annotators is 0.544, higher than the mean human-to-human correlation of 0.462. Adding it as a sixth coder nonetheless *lowers* ordinal α from 0.325 to 0.300: it tracks the consensus without belonging to the distribution. Against its own five samples it is far more self-consistent than the humans are with each other ($\alpha = 0.689$ versus 0.325), and averaging the samples is not cosmetic, since correlation against the human mean climbs from 0.611 at $K = 1$ to 0.719 at $K = 5$. Its rubric-level explanations show the same pattern: on identical pairs it reports an error in 0.0% of samples, and on LEXFLIP it reports some error in 78.1% and names *incohérence*, the correct type for a meaning-reversing edit, in 54.6%. Of six rules for merging the judge with bidirectional NLI, every calibration fitted on the training split alone and the diagnostic never entering the fit, none beats both parents on both axes; the best, a weighted blend at $w = 0.74$, reaches $r = 0.762$ and margin 0.574, gains smaller than the redraw-to-redraw variation. We report this as a negative result rather than proposing a metric.

4.6. What Training Against the Protocol Buys

Table 8 reports the grid of Section 3.5, and the first result is a negative one: no training target moves the correlation by more than the scalar baseline's own spread across seeds (± 0.082), which is

Table 6. Calibrated comparison on the FRJUDGE test split (89 items), mean over 10 seeds. RMSE_{raw} is the decimal-scaling protocol in current use; RMSE_{cal} fits an isotonic map on the training split. r and ρ are both computed after calibration; being monotone, the map leaves ρ almost unchanged (at most 0.014) but shifts r by up to 0.074.

Metric	r	ρ	RMSE_{raw}	RMSE_{cal}	over%
Unsupervised, French-capable					
BERTScore-CamemBERTv2	$0.291_{\pm 0.09}$	0.326	2.62	2.36	49.3
BERTScore-CamemBERTv2-Recall	$0.352_{\pm 0.12}$	0.398	2.59	2.30	47.2
BERTScore-FlauBERT	$0.482_{\pm 0.08}$	0.459	2.16	2.12	49.4
LaBSE	$0.409_{\pm 0.06}$	0.402	2.33	2.21	47.2
Sentence-CamemBERT	$0.296_{\pm 0.05}$	0.287	2.68	2.31	49.3
Para-mUSE-mpnet	$0.252_{\pm 0.08}$	0.266	2.88	2.35	47.2
mE5-base	$0.358_{\pm 0.08}$	0.356	2.52	2.26	48.0
NLI-fwd (no hallucination)	$0.368_{\pm 0.06}$	0.352	3.85	2.25	49.3
NLI-bwd (no omission)	$0.340_{\pm 0.09}$	0.355	3.70	2.28	46.9
NLI-bidirectional (min)	$0.395_{\pm 0.07}$	0.405	3.74	2.22	46.6
Trivial surface features					
$-\lvert\lvert s\rvert - \lvert o\rvert\rvert$ (length difference)	$0.641_{\pm 0.04}$	0.624	3.25	1.85	47.0
Token Jaccard	$0.303_{\pm 0.06}$	0.331	3.34	2.34	48.0
Length ratio	$0.580_{\pm 0.05}$	0.560	3.43	1.97	49.1
Surface-Ridge (10 features, supervised)	$0.621_{\pm 0.05}$	0.588	1.89	1.89	45.8
Prompted LLM judge					
LLM-judge (deepseek-chat, k=5)	$0.755_{\pm 0.05}$	0.753	2.93	1.57	43.9
Human ceiling (one annotator)	**0.597**	0.620	–	3.27	–

Table 7. Williams tests against the trivial length baseline $\ell = -||s|-|o||$, which reaches $r = 0.613$ with the human label h over all 297 items. $r(m,\ell)$ is each metric's correlation with the length feature itself, the dependence the test corrects for. Positive t favours ℓ.

Neural metric m	$r(m,h)$	$r(m,\ell)$	t	p
BERTScore-CamemBERTv2	0.344	0.349	5.05	<0.001
BERTScore-CamemBERTv2-Recall	0.396	0.443	4.43	<0.001
BERTScore-FlauBERT	0.484	0.389	2.61	0.010
LaBSE	0.424	0.362	3.66	<0.001
Sentence-CamemBERT	0.222	0.148	6.33	<0.001
Para-mUSE-mpnet	0.225	0.159	6.32	<0.001
mE5-base	0.318	0.316	5.40	<0.001
NLI-fwd (no hallucination)	0.320	0.388	5.66	<0.001
NLI-bwd (no omission)	0.307	0.258	5.36	<0.001
NLI-bidirectional (min)	0.365	0.350	4.68	<0.001

itself far smaller than the difference of roughly a quarter of a correlation point that an 89-item split can resolve. The differences that survive are in the other columns. The distributional head is markedly steadier than the scalar head it replaces (± 0.029 against ± 0.082 over the same five seeds), reaches the best RMSE in the table (1.99), and roughly doubles the rank correlation ($\rho = 0.578$ against 0.280), which is what one expects from a target carrying the shape of the rating distribution and not only its mean. The quantile head does precisely what it was asked to do: it overshoots the human label on 9.9% of test items against 25.4% for the scalar head, the lowest rate of any model here. The risk asymmetry a legal setting calls for can therefore be written into the loss rather than hoped for, and it costs nothing in correlation.

Trained constraints do not cross a change of document. Every model in Table 8 scores 0% on the identical-pair check, the two augmented ones included, although identical pairs are in their training mixture; the augmented models pass the unrelated check on 77% and 69% of probes and the un-augmented models on none. The asymmetry is the one our probe design predicts. Our unrelated probes are drawn from the same two statutes the training mixture draws its unrelated pairs from, so passing that check is an in-distribution result; our identical probes are statutory sentences while the mixture's identical pairs are FRJUDGE clauses, so passing that one would require generalising across source documents, and no model does. The same holds for the diagnostic: DA+LEXFLIP, trained on minimally-edited negatives, reaches a margin of 0.051 against 0.670 for untrained bidirectional NLI, its negatives generated from FRJUDGE originals and its probes from statutes. Training on a constraint buys satisfaction of it in distribution and nothing at all across a change of source. The same separation explains the augmented mixtures' correlations, which are the clearest effect in the table: DA loses 0.350 against the un-augmented baseline and DA+LEXFLIP 0.420, on five seeds out of five in both cases, and DA varies by ± 0.254 across seeds, reaching $r = -0.188$ on one of them.

One column should be read with care. LEXFLIP false-acceptance is the share of flipped pairs still scored ≥ 7, so it is confounded with where a model puts its predictions overall. The distributional head's 69% reflects a head that scores high in general (its identical probes average 7.22), and the 0% of the scalar heads reflects models that score low on everything, not models that reject the edit.

4.7. What Survives a Change of Document

A metric meta-evaluation that never changes source document cannot distinguish legal sensitivity from familiarity with a form. Random splitting lets clauses from the same insurance form appear in both training and test, and insurance forms are highly templated, so we rerun the scalar baseline and the two augmented mixtures on a document-grouped split in which no source document straddles the boundary, size-matched to the random split pair for pair. The remaining four configurations differ from the baseline in what they are trained to predict rather than in what they can exploit about a source form, so the leakage they are exposed to is the same.

The comparison does not resolve. The scalar baseline's correlation falls from 0.542 on the random split to 0.454 on the grouped one, a drop of 0.088 that is in the direction leakage would predict, and it does not survive inspection: the per-seed differences are $+0.169$, -0.097, -0.416, -0.182 and $+0.086$, so two of the five seeds *improve* under the harder split and the grouped spread (± 0.143) is wider than the gap itself. The two augmented mixtures move by -0.002 and $+0.026$, which is to say not at all. As a control, the untrained NLI reference is fitted to nothing and should be indifferent to how the corpus is partitioned; it moves from 0.395 to 0.391, confirming that the two splits are otherwise comparable and that the variance above is the models' and not the split's.

We therefore do not claim that document-level leakage inflates these correlations, and we hold our own evidence to the standard Section 6 applies elsewhere. What we can say is narrower and, we think, more useful to anyone planning a successor corpus: FRJUDGE is too small and too concentrated to answer the question. Its 297 pairs come from only 21 source forms and the three largest hold 71% of them, so the largest form alone is 37% of the corpus, larger than the 30% test quota, and always falls in training. The grouped test split is therefore always drawn from the 188 pairs that form leaves behind, at which size the seed-to-seed variation swamps the effect being measured. A corpus intended to support an out-of-domain claim needs its documents spread far more evenly, and this is the aspect of the design we would change first.

4.8. Stratifying by Length Change

If a metric's correlation is concentrated in the items where the simplification is much shorter, it is detecting deletion rather than legal change. Table 9 reports Pearson r inside each tercile of $||s|-|o||$. Within a tercile the length signal is largely held constant, so a metric that retains its correlation there is using something other than deletion; one whose correlation collapses was reading length all along. This is a weaker form of the same dissociation logic as Section 3.1: it holds a confound approximately fixed by conditioning rather than exactly fixed by construction.

The result qualifies Section 4.4 in a way that cuts against our own argument, and is worth stating carefully. The length feature's correlation is overwhelmingly a *between*-tercile effect: pooled over the corpus it reaches 0.613, but inside the small and medium terciles it falls to 0.243 and 0.165. Once items of comparable length change are compared with each other, knowing the length change tells you little. It recovers to 0.585 only in the tercile where the simplification is much shorter than the original, which is exactly where deletion is available to be detected. The neural baselines have the opposite profile: in the medium tercile, where the length feature retains almost nothing, BERTScore over FlauBERT holds 0.449 against a pooled 0.484 and LaBSE rises to 0.532 against a pooled 0.424. The supervised surface ridge tracks the raw length feature bucket for bucket, which is what one would expect if it is largely a length model with

Table 8. Trained metrics, mean over 5 seeds, evaluated on the human-annotated test split only. *ident.* and *unrel.* are the two sanity checks in current use; LEXFLIP false-acceptance is the share of legally flipped pairs still scored ≥ 7 ("accurate" under the published rubric). The margin is the quantity defined in Section 2. It is shown as '–' where a model places the identical and unrelated probes within one point of each other, because a model with no working range has none to spend on the edit.

Model	r	ρ	RMSE	over%	ident. pass	unrel. pass	LEXFLIP false-acc.	LEXFLIP margin
JudgeBERT-Scalar (baseline)	$0.542_{\pm 0.08}$	0.280	2.48	25.4	0	0	0	–
JudgeBERT-Dist (soft labels)	$0.572_{\pm 0.03}$	0.578	1.99	42.9	0	0	69	0.015
JudgeBERT-Annot (5 heads)	$0.534_{\pm 0.05}$	0.585	3.47	12.6	0	0	0	–
JudgeBERT-Quantile (tau=.25)	$0.553_{\pm 0.06}$	0.377	3.28	9.9	0	0	0	–
JudgeBERT-MT (+charact.)	$0.519_{\pm 0.06}$	0.232	2.51	26.3	0	0	0	–
JudgeBERT-DA	$0.192_{\pm 0.25}$	0.112	2.31	42.5	0	77	18	0.019
JudgeBERT-DA+LexFlip	$0.122_{\pm 0.10}$	0.174	2.60	35.3	0	69	0	0.051
NLI-bidirectional (untrained)	0.395	0.405	2.22	46.6	1	–	–	*0.670*
Human ceiling	**0.597**	0.620	3.27	–	–	–	–	–

Table 9. Correlation with the human label *within* terciles of the absolute length change, pooling test predictions over seeds. Restricting to a tercile removes most of the length variance; a metric that measures legal meaning should degrade far less than one that measures deletion.

Metric	r within tercile small	medium	large	all
$-\lvert\lvert s\rvert - \lvert o\rvert\rvert$	0.243	0.165	0.585	0.613
BERTScore-FlauBERT	0.198	0.449	0.413	0.484
LaBSE	0.174	0.532	0.240	0.424
NLI-bidirectional	0.058	0.269	0.291	0.365
Surface-Ridge	0.118	0.305	0.540	0.620
JudgeBERT-Scalar	0.042	0.149	0.602	0.496

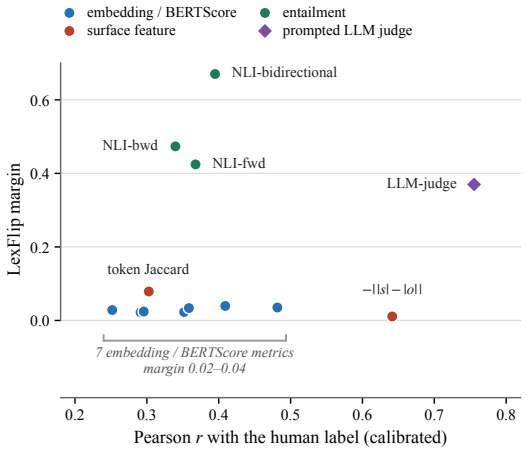
extra terms. The small tercile is hard for everything, which we read as a floor effect rather than as a result: when the two sentences are close to the same length, none of these signals has much to go on.

The honest reading is therefore narrower than "legal meaning preservation is length preservation". The corpus-level ranking is dominated by length, and any comparison that omits a length control will attribute to a metric what it should attribute to that confound. But the neural metrics are not *only* reading length. Where the length signal is weakest, they are the ones that hold up. Whatever they are reading in that stratum, it is not the length change. What Section 4.1 adds is that the residual signal they retain is still not legal sensitivity, since the same metrics move almost not at all when the law is changed and the length is held exactly fixed. The two analyses answer different questions, and a metric has to survive both.

4.9. Correlation and Legal Sensitivity Come Apart

Over the 13 scorers carrying both a calibrated r and a margin the rank correlation between them is $\rho = 0.24$ (95% interval $[-0.39, 0.75]$), an absence of resolution rather than a demonstrated null; the extremes are the durable part. The length feature outscores every semantic metric and has the lowest margin of anything we test (0.011), while bidirectional NLI sits mid-table on correlation and alone reacts to the edit. The empty upper-right quadrant of Figure 4 is the finding: nothing we test is both a good predictor of the human label and sensitive to a legally decisive edit.

Figure 4. The two orderings come apart. Horizontally, calibrated correlation with the human label, the quantity a conventional comparison selects on; vertically, the share of working range spent on an edit that reverses the law.



5. Discussion

A metric claiming to measure legal meaning should report three checks, not two: identical pairs (saturation), unrelated pairs (bottoming out), and LEXFLIP (does it move when, and only when, the law moves?). The first two are jointly satisfiable by lexical overlap; the third is not, so a low margin is a fixable deficiency rather than a limit of automatic evaluation. Only legal sensitivity resists calibration: rank on it, and build the next metric on a repaired entailment objective.

The practical consequences fall into three places. For anyone *building* a metric, the diagnostic says that the objective and not the encoder is what needs changing. Seven configurations sharing an encoder and differing only in target and loss move the correlation by less than seed noise, while an untrained bidirectional entailment score, which is the only scorer here whose training objective is about directional meaning containment rather than similarity, outperforms all of them on the one axis that matters. For anyone *reporting* a metric, R2 and R3 are cheap and change conclusions: isotonic calibration erases most of the RMSE gap that uncalibrated

tables attribute to supervision, and a length control moves a semantic metric from "competitive with humans" to "below a feature that counts words". For anyone *deploying* a metric, the asymmetry the prompted judge shows is the sharpest warning in the paper. It penalises a clause that becomes more permissive far more than one that becomes more demanding, which is a defensible policy and a disastrous default if it is not the policy anyone chose.

Two findings generalise beyond legal text. The first is that a validation protocol built only from endpoint checks can be satisfied by construction, and the repair, holding one variable fixed while moving the other, applies to any evaluation where the property of interest is correlated with surface form in naturally occurring data. The second is that training against a diagnostic buys satisfaction of it in distribution and nothing across a change of source document, which is a reason to keep such diagnostics out of training mixtures and to treat them, as we do, as necessary conditions rather than benchmarks to optimise. Both point at the same practical recommendation for whoever builds the next corpus in this area. Source documents should be spread evenly enough that a grouped split is possible, annotator identity should survive into the public release so that a ceiling can be measured downstream, and a held-out set of minimally edited negatives should be published alongside the ratings and kept out of every training mixture, so that legal sensitivity can be reported next to correlation rather than inferred from it.

6. Limitations

Our LEXFLIP labels are definitional rather than annotated: we assert that replacing *doit* with *peut* changes legal meaning rather than having lawyers rate each perturbation. That is defensible for tier-1 edits, reported separately from tier 2, but a small expert study would be stronger. The edits are single and template-generated, so a system could be tuned to them without acquiring general legal sensitivity, which is why the diagnostic is a necessary condition and not a benchmark to optimise.

The same caveat bears on our best result. Our perturbations are largely negation, modality, quantifier and antonym contrasts, close to what NLI training data is built around, so bidirectional NLI's margin of 0.670 may reflect familiarity with their *form* rather than sensitivity to their legal consequence. Yang et al. [19] give the general version: a counterfactual edit isolates the variable of interest only against a meaning-preserving paraphrase control, which we do not run and the next version of the diagnostic should carry.

Finally, the diagnostic and the corpus comparison run on different text, insurance forms against two statutes, and answer different questions; the judge results rest on one model under a rubric reconstructed from the published description, a load-bearing assumption given that wording alone moves the margin by 0.267; and the ceiling is computed on all 297 items while model correlations run on 89-item splits, so "model $>$ ceiling" carries the sampling uncertainty of both. The corpus itself is one language and one legal system, and whether the same dissociation holds for common-law English drafting is an open question our design cannot answer.

7. Conclusion

Legal meaning preservation metrics are validated today by two checks that lexical overlap satisfies by construction. We proposed a third that it cannot, released LEXFLIP to administer it, and found that every similarity-based metric in current use spends between 0.022 and 0.039 of its own working range on an edit that reverses a statutory obligation. Bidirectional NLI is the exception at 0.670, and it is the family the identical-pair convention would have excluded. On the only annotated corpus for this task a supervised ridge over word counts matches a measured human ceiling and beats every semantic metric, and a prompted judge exceeds the ceiling while awarding a reversed clause 6.69 out of ten. Correlation with human ratings and sensitivity to a change in the law are, on the evidence here, close to independent. Reporting both is the smallest change that would make the difference visible, and repairing the entailment objective is where we would look for the next metric.

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Ethical Statement

This study does not contain any studies with human or animal subjects performed by the author. It reuses an existing annotated corpus, FRJUDGE, whose five annotators are pseudonymised as A–E throughout; our released code pseudonymises annotator identifiers on load, and we ship the pseudonymisation script for anyone redistributing a reconstruction.

Beauchemin et al. [3] caution that their metric informs a practitioner about legal meaning preservation rather than supplying a complete juridical analysis, and our results sharpen that into a rule. A metric that cannot distinguish *doit* from *peut* must not gate whether a simplified clause is safe to publish, and the appropriate deployment is triage, the ranking of clauses for human review, rather than certification. Nothing in this article constitutes legal advice. The perturbed sentences we release are deliberate misstatements of Quebec law, labelled as such in the distribution, and are not statements of the law.

Conflicts of Interest

The author declares that he has no conflicts of interest to this work.

Data Availability Statement

The data and code that support the findings of this study are openly available at `https://github.com/Nyvora-Vision-Labs/LEXFLIP`. The repository contains the LEXFLIP generator and the released item set, the scoring harness, the reconstruction of FRJUDGE from its raw export (which recovers annotator identity and source document, both dropped from the public JSONL), the document-grouped splits, and the result files behind every table in this article. FRJUDGE itself is distributed by Beauchemin et al. [3] and is not redistributed here.

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